

MALMI WITHANAGE

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Associate Software Engineer specializing in robust backend systems, secure multi-tenant isolation, and data pipelines. Experienced in building event-driven worker layers and shipping production-grade, peer-reviewed machine learning applications. Committed to rigorous backend test coverage, data consistency, and end-to-end service deployment.

TECHNICAL SKILLS

- **Languages:** TypeScript, JavaScript, Python, Java, C++, HTML5, CSS3
- **Frontend:** React, Next.js, React Native (Expo), Tailwind CSS, State Management
- **Backend Architecture:** NestJS, FastAPI, Node.js, REST APIs, Microservices, JWT, OAuth 2.0, RabbitMQ
- **Data & Machine Learning:** PostgreSQL, MongoDB, Redis, Scikit-learn, XGBoost, Random Forest, SARIMA, Pandas, NumPy
- **Cloud & DevOps:** AWS (EC2, RDS), Microsoft Azure, Docker, Cloudflare, Git, GitLab CI/CD, BullMQ

PROFESSIONAL EXPERIENCE

Associate Software Engineer | Lanka Communication Services (Pvt) Ltd Oct 2025 – Present

- Architected and expanded core features for an enterprise Learning Management System (LMS) using NestJS, Next.js, RabbitMQ, and MongoDB to meet strict delivery timelines.
- Designed and shipped a complex timetable scheduling module, drastically reducing post-deployment bugs and hotfixes through rigorous local testing with Docker and Redis cache pipelines.
- Mentored engineering interns, led daily Agile standups, and drove cross-project issue resolution to stabilize legacy codebases and accelerate team onboarding velocity.

Trainee Associate Software Engineer | Lanka Communication Services (Pvt) Ltd Jul 2025 – Oct 2025

- Accelerated full-stack feature development for a high-traffic LMS platform, optimizing message queues with RabbitMQ and data layers with MongoDB.
- Refactored core scheduling logic and timetable sub-systems, actively resolving blocking repository issues to preserve system stability across multiple active company projects.

Junior Software Engineer | 2xDev Mar 2025 – Jun 2025

- Owned the resolution of high-priority web application bugs and actively contributed to architectural decision-making processes for scaling core digital products.
- Engaged in rapid Agile delivery frameworks and daily engineering standups to coordinate hotfixes and feature releases alongside cross-functional team members.

Software Engineer Intern | 2xDev Jun 2024 – Dec 2024

- Developed scalable REST API endpoints and interactive user interface components using modern JavaScript/TypeScript and web frameworks.
- Maintained local Docker environments and containerized configurations to ensure absolute parity across development, testing, and staging lifecycles.

TECHNICAL PROJECTS

Logi Track — Multi-Tenant Logistics System (NestJS, PostgreSQL, Prisma, Redis, BullMQ, Supabase)

- Enforced absolute tenant data isolation utilizing a custom cryptographic JWT Guard layer, preventing access parameter tampering.
- Engineered an asynchronous CSV export pipeline using BullMQ and Redis, offloading heavy computations from the synchronous request thread.
- Implemented automated data isolation verification using Jest integration suites, covering complex cross-tenant rejection tests.

CinoGrow — ML Agriculture Decision Support System (FastAPI, Python, XGBoost, SARIMA, React Native, Docker, AWS)

- Built dynamic distillation time regression models (R^2 0.970) and oil-quality classification loops using 5-fold cross-validation.
- Designed a SARIMA time-series model (3.85% MAPE) to predict volatile seasonal price fluctuations for smallholder farmers.
- Containerized the entire application stack via Docker, enabling seamless automated deployment to production clusters on AWS.

ACADEMIC PUBLICATION

18th ECAI 2026 International Conference on Electronics, Computers and Artificial Intelligence (Bucharest, Romania)
"CinoGrow: An Integrated Machine Learning Decision Support System for Precision Cinnamon Farming" Co-authored with SLIIT research cohort. Supervised by Ms. S. Silva & Ms. N. Gamage. DOI: 10.1109/ECAI69016.2026.11613746

EDUCATION

BSc (Hons) in Information Technology | Sri Lanka Institute of Information Technology (SLIIT) 2022 – 2026